



Collaborative Innovation in Audio Description: Integrating Community, Education, and AI to Advance Accessible Media

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YouDescribe is a living innovation lab

- 13 years of collaboration driving YouDescribe's evolution
- Built through partnerships with Smith-Kettlewell Eye Research Institute, San Francisco State University, and Northeastern University
- Powered by students, researchers, developers, and the accessibility community
- Continuous co-design: learning, rebuilding, and improving together (2013 → 2017 → 2019 → 2024 → 2026)
- Blending research, education, and real-world impact to advance accessible media



What is YouDescribe.org?

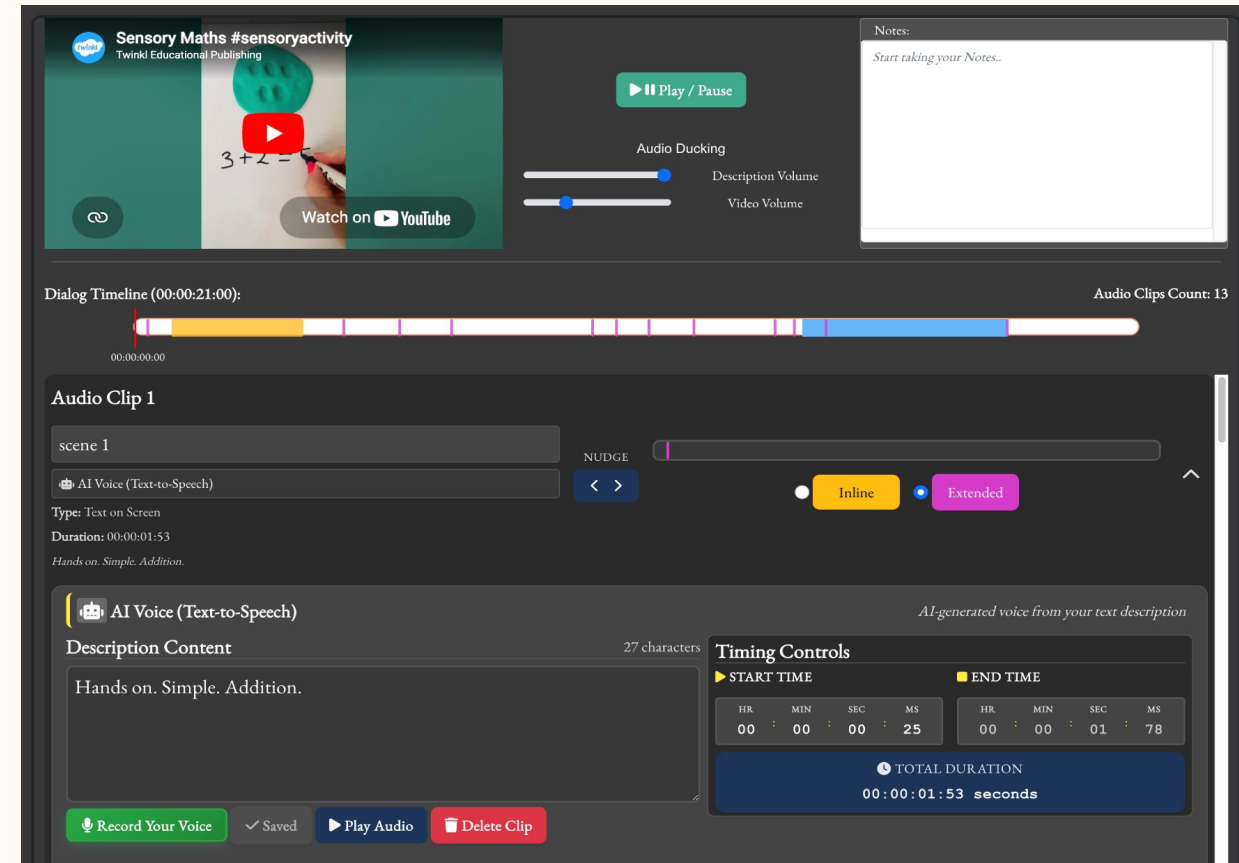


- It's a free accessibility tool to add audio description to YouTube videos built especially for the BLV community
- Viewers request videos they would like described through a wishlist
- Volunteers create a script, record tracks, and edit track placement
- The tracks and timing data is stored in a separate API and played together with YouTube video

What YouDescribe isn't

YouDescribe isn't a replacement for professional audio description. It's a tool for what would be left undescribed due to:

- Time
- Cost
- Importance



What data do we collect?

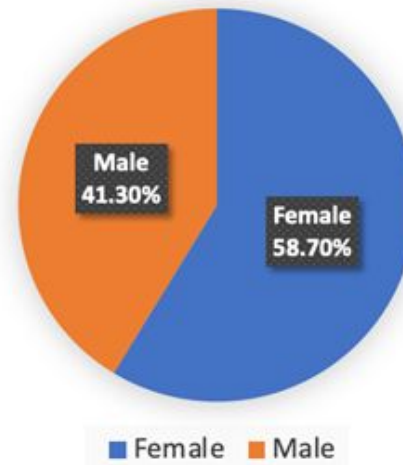
(and don't collect)

- Our viewer and describer privacy are very important to us.
- All data is anonymized
- No data is collected for visitors to www.youdecribe.org that are under 18 years of age
- We do not share email addresses, names, or any other identifying information
- General visitor data is collected by google analytics (started in 2020)

Visitors by Country



Visitors by Gender



Top Channels



You Described, We Archived Data Repository

URL : <https://github.com/youdescribe-sfsu/You-Described-We-Archived>

Steps to download the data

- a. You will be asked to register
- b. You will be given an API key
- c. Run the provided Python file (download script)

You Described, We Archived © 2022 by Smith-Kettlewell Eye Research Institute, San Francisco State University is licensed under [CC BY-NC-ND 4.0](#).

Is the YuWa data any good?

Oxford University Group QuerYD

Dataset trainer for retrieval and event localization in videos

Compared to other training data sets content descriptions made with YouDescribe data:

- more relevant than dialogue alone
- more detailed than previous description attempts using crowd-sourced image descriptions

Our YuWA Dataset

Our data set is better

We have more metadata

We have twice the audio (both the video dialog and the audio descriptions)

We have extended descriptions

Better metadata means more research applications

Past Research: AI-Generated Drafts to Help Novice Describers

- In 2019, we developed a human-in-the-loop Machine Learning approach to video description by automating video text generation and scene segmentation and allowing humans to edit the output
- This approach helps produce high-quality descriptions while keeping the barriers to entry low for volunteer describers
- The system was significantly faster and easier to use for first-time video describers as compared to a human-only control condition with no assistance
- Blind and low vision users rated the quality of the descriptions and their understanding of the topic more highly for the AI-assisted descriptions than for the descriptions created in the human-only condition

Past Research: The Potential of a Tandem System for Audio Description

YouDescribe's user interface was tested in 2022 with 26 blind and low vision end users

Users watched videos under six conditions:

1. No tools
2. Visual dialogue agent (Q&A) only
3. AI-generated baseline description only
4. Human-revised baseline description only
5. AI-generated baseline descriptions + Q&A
6. Human-revised baseline descriptions + Q&A

Recorded use & assessed self-reported comprehension, enjoyment, and overall satisfaction with video watching experiences



Past Research: The Potential of a Tandem System for Audio Description

- Only 50% of the visual dialogue agent's responses to user questions were inaccurate (AI was pretty bad in 2022). Users indicated that the agent alone didn't significantly improve their comprehension or enjoyment of the video
- Users tended to activate the Q&A when the baseline description was incorrect or sparse
- AI-generated and human-revised descriptions both had a similar positive impact on users' ability to understand what happened in a video, but they enjoyed the human-revised descriptions significantly more
- Surprisingly, when AI-generated baseline description and the visual dialogue agent were used in tandem, participants reported similar levels of enjoyment and comprehension as when they were given human-revised descriptions

Past Research: The Potential of a Tandem System for Audio Description

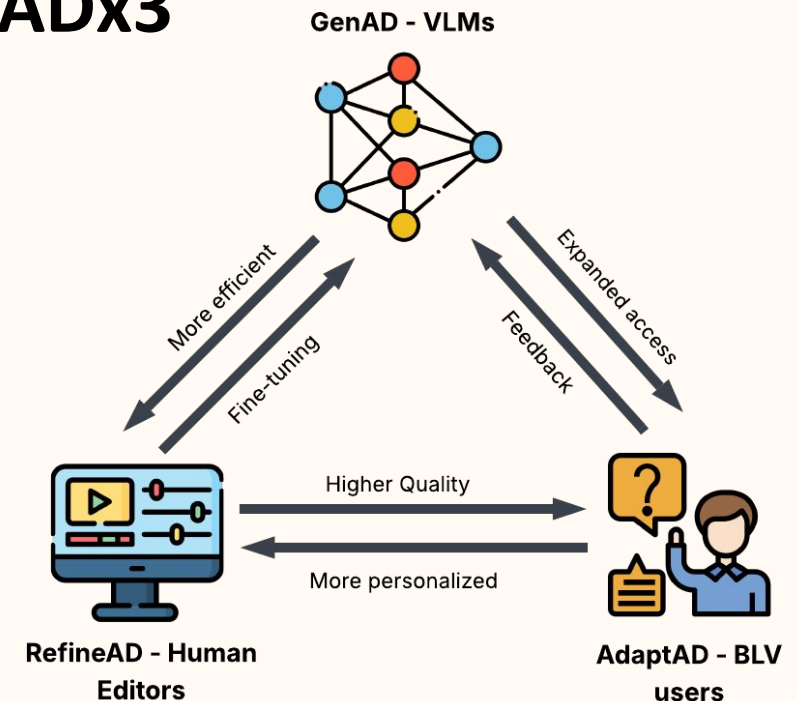
- AI's visual identification capabilities have improved a lot since 2022
 - Accuracy is much higher (estimate >90%)
 - AI audio description still lags behind human-revised description in other aspects, and requires a great deal of prompt engineering to produce better quality descriptions
- We're planning another blind and low vision end user study in 2026

AI-Assisted Audio Description: Technology and Research

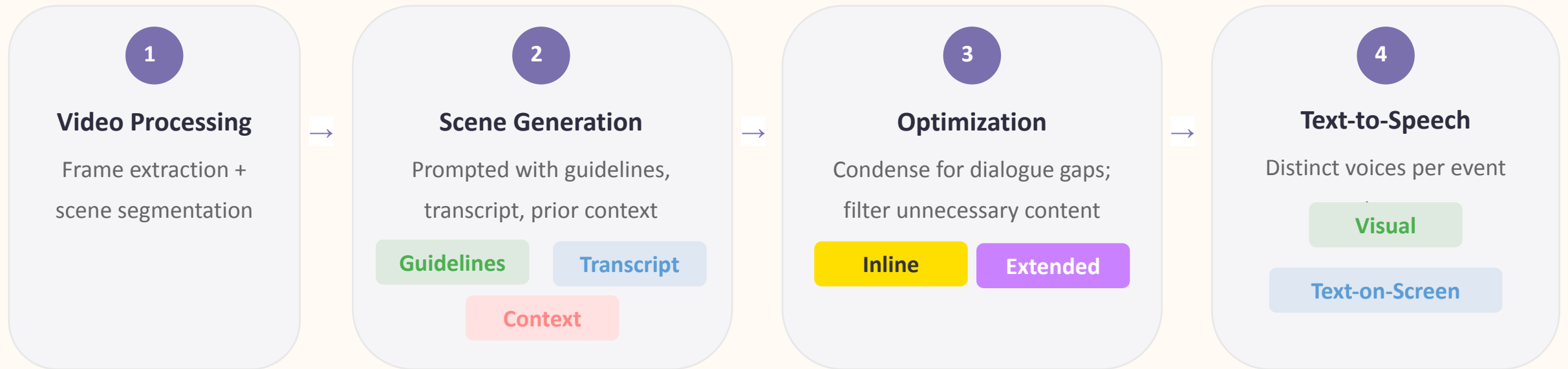
- **Motivation**

Our past work found that AI baseline descriptions paired with a visual dialogue agent gave BLV users enjoyment and comprehension comparable to human-revised AD, combining all three components can support both volunteers and BLV users on YouDescribe

- **ADx3**



GenAD - AI generated AD Pipeline



GenAD - AI generated AD Pipeline

Grounding in professional AD guidelines and context produces concise, character-aware descriptions, not verbose, caption-like outputs.



- × **Naive output (without guidelines → overly detailed):** A young woman with brown hair tied back stands in a vast desert landscape. She is wearing a beige, sleeveless outfit with a brown belt and arm wraps. The woman looks determined as she gazes into the distance. Her expression is focused. The woman reaches for a cylindrical object attached to her belt.
- **With guidelines (concise but generic):** A woman stands in a vast desert, dressed in a beige outfit. She is scanning the horizon, cylindrical object attached to her belt.
- ✓ **With guidelines + context (concise and specific):** **Rey** stands in a vast desert, scanning the horizon, lightsaber hilt at her waist.



Natural Gap

First pass: A lush, expansive forest stretches across rolling hills under a bright blue sky dotted with fluffy white clouds. (Extended)

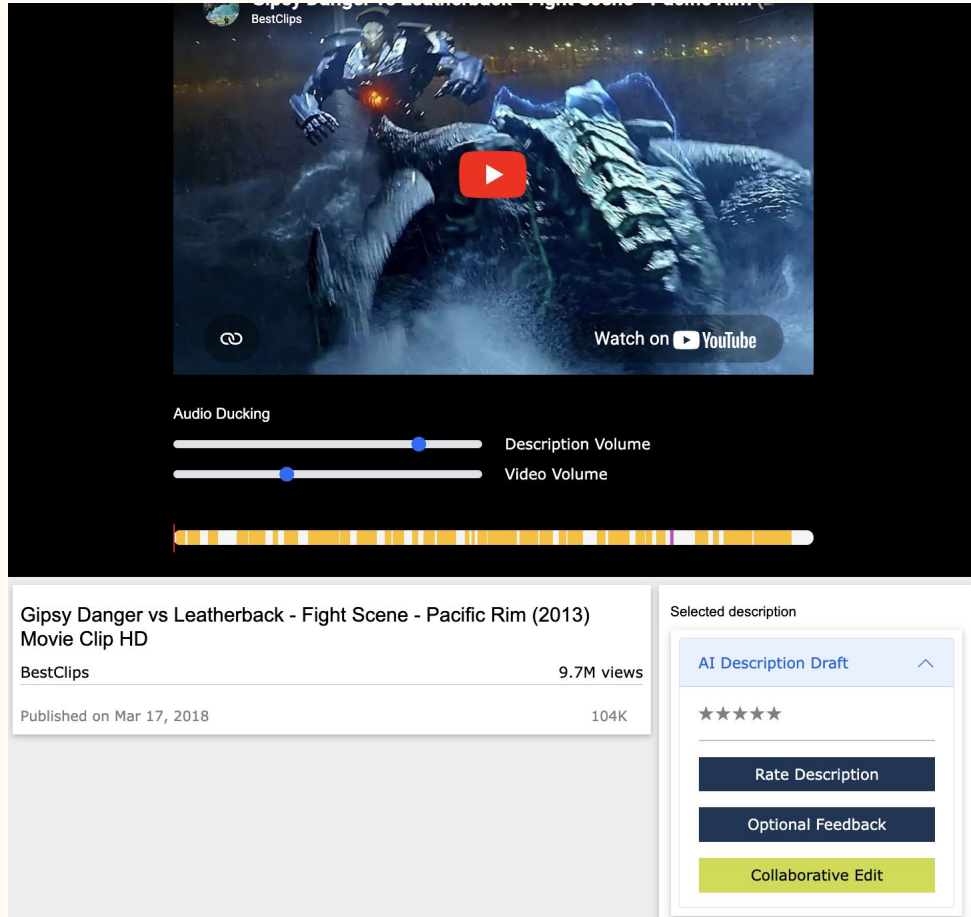
✓ **Optimized:** Lush forest covers rolling hills, sky dotted with clouds. (Inline)

Natural Gap

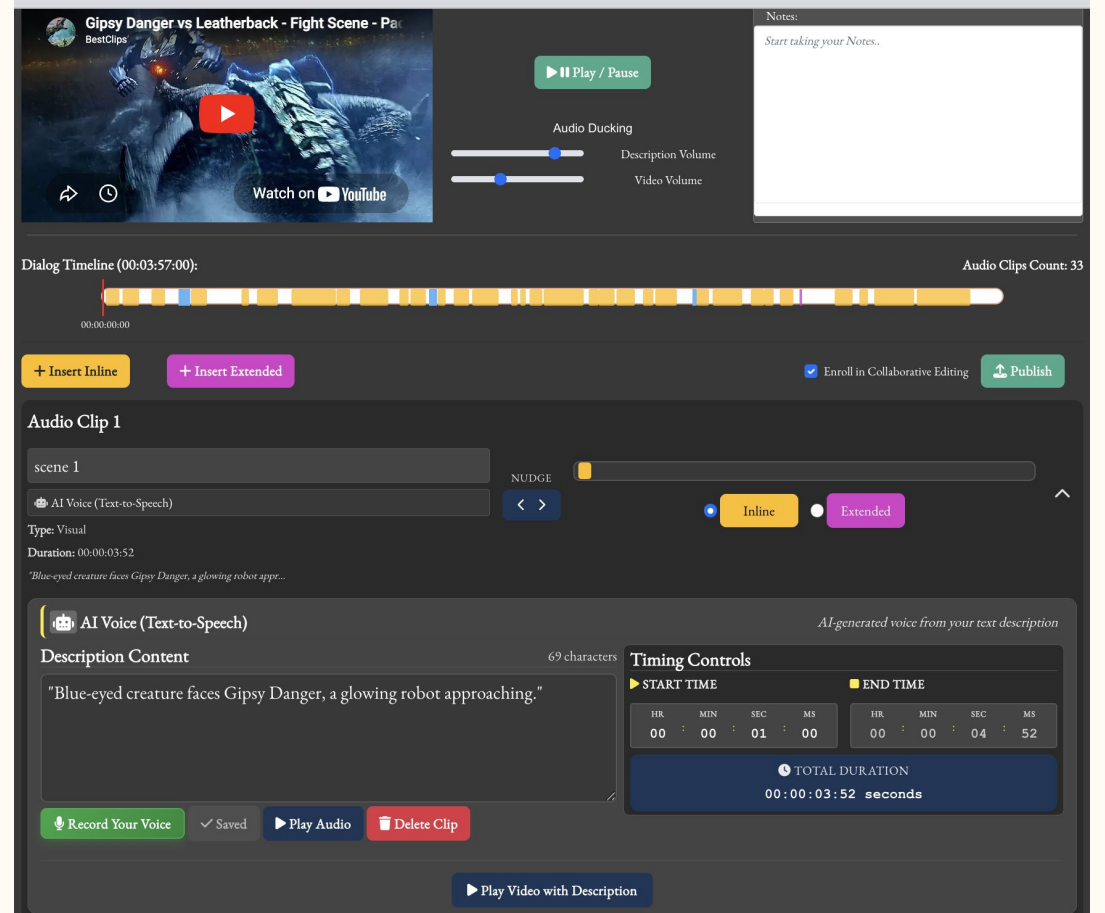
First pass: A lush green tree canopy fills the frame, with leaves gently swaying in the breeze. The sky is clear and bright. A chimpanzee appears, partially obscured by the foliage. It moves its arms, possibly reaching for something or interacting with its environment. (Extended)

✓ **Optimized:** Green canopy sways. Chimpanzee moves, arms reaching through leaves. (Inline)

GenAD -> RefineAD



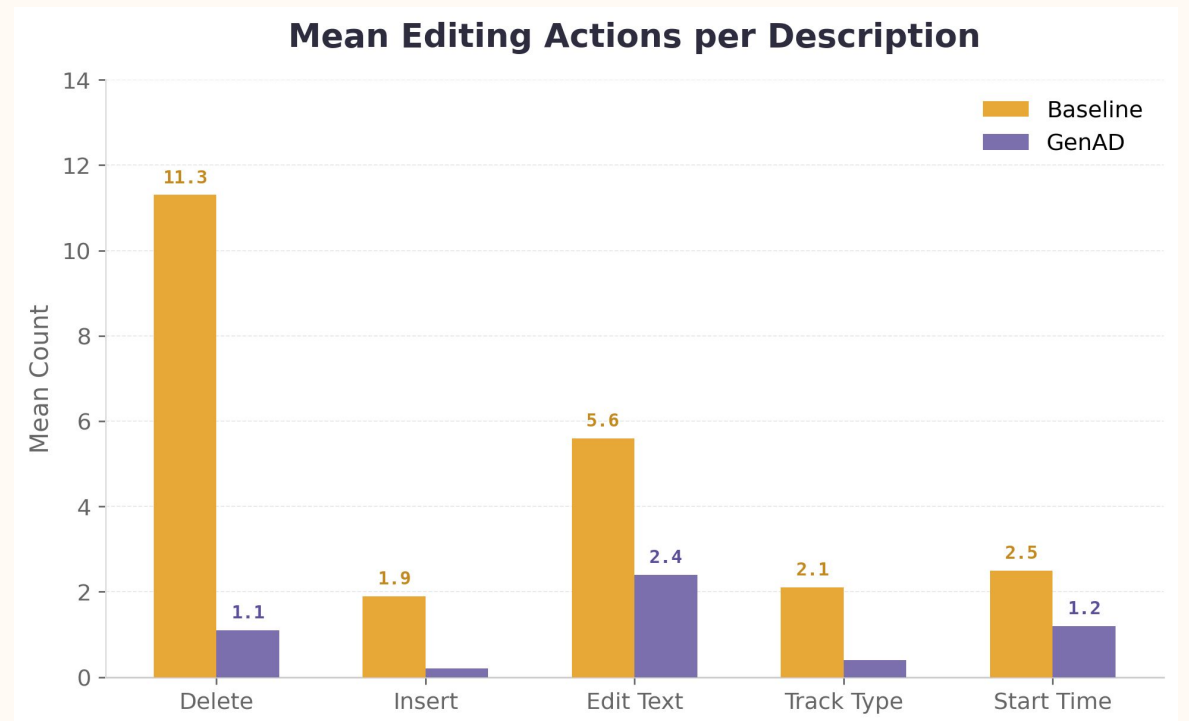
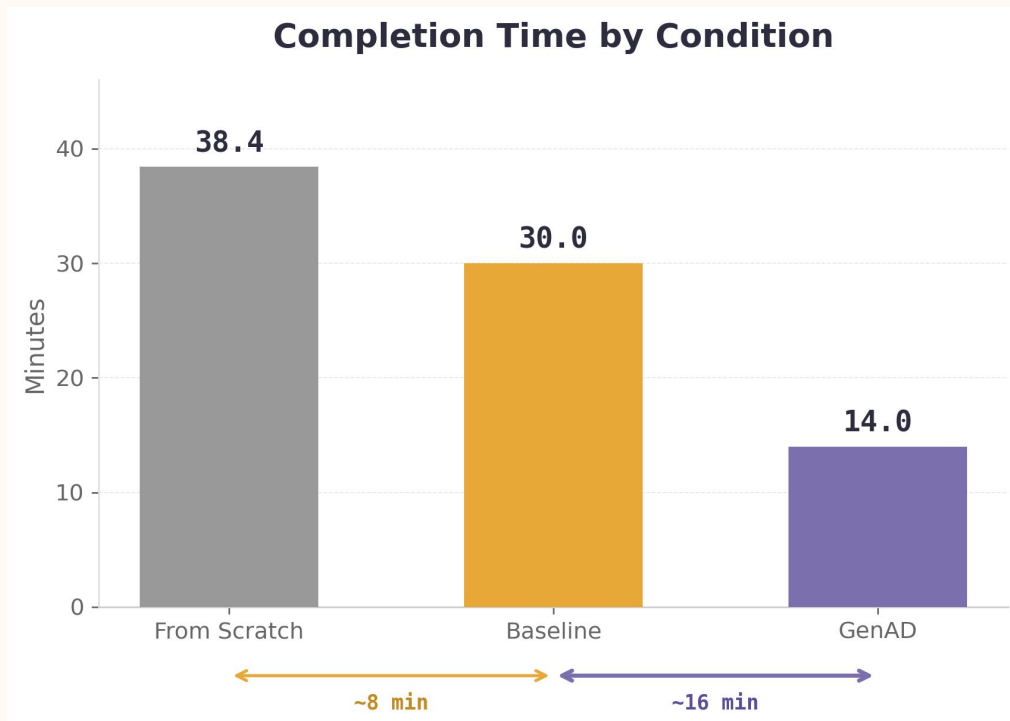
The GenAD interface displays a video player at the top with a red play button and a "Watch on YouTube" link. Below the player are sliders for "Audio Ducking", "Description Volume", and "Video Volume", followed by a progress bar. The bottom section contains the video title "Gipsy Danger vs Leatherback - Fight Scene - Pacific Rim (2013) Movie Clip HD", the channel "BestClips", view count "9.7M views", and publication date "Published on Mar 17, 2018". A "Selected description" panel on the right shows an "AI Description Draft" with five stars, a "Rate Description" button, an "Optional Feedback" button, and a "Collaborative Edit" button.



The RefineAD interface provides a detailed editing and AI generation environment. It features a video player at the top with "Play / Pause", "Audio Ducking", "Description Volume", and "Video Volume" controls, along with a "Notes" section. Below the player is a "Dialog Timeline (00:03:57:00)" with a progress bar and "Audio Clips Count: 33". The interface includes buttons for "+ Insert Inline" and "+ Insert Extended", and checkboxes for "Enroll in Collaborative Editing" and "Publish". The "Audio Clip 1" section shows a text input field with "scene 1", a "NUDGE" slider, and "AI Voice (Text-to-Speech)" controls. The "Description Content" field contains the text "Blue-eyed creature faces Gipsy Danger, a glowing robot approaching." with a character count of 69. The "Timing Controls" section includes "START TIME" and "END TIME" fields, a "TOTAL DURATION" of 00:00:03:52 seconds, and buttons for "Record Your Voice", "Saved", "Play Audio", and "Delete Clip". A "Play Video with Description" button is at the bottom.

RefineAD - Human refinement

We did an user study with 30 novice describers to refine Baseline and GenAD drafts



RefineAD - Human refinement

- AI drafts act as scaffolding novices start from a structure instead of a blank page, shifting from author to editor.

“Having an initial draft helped me focus on accuracy, timing, and clarity rather than starting from a blank page.”

- Weak AI drafts cost more to fix than to rewrite, novices get anchored to flawed phrasing and lose creative freedom.

“...I spent quite a lot of time correcting it instead of having the time to describe it...”

AdaptAD - Description on Demand



—Key 'D'—>

✗ **Without strategic prompting** → response tends to be long and redundant

The scene takes place in a snowy landscape with a line of pine trees in the background. A single, small purple crocus is blooming in the snow in the foreground. A snowman stands in front of the same crocus, looking down at it with a happy expression. Footprints in the snow lead to him. The general composition remains the same, but the lighting is significantly different.

✓ **With strategic prompting** → response is concise and contextually grounded.

Olaf the snowman smiles at a small purple crocus blooming in the snowy forest.



—Key 'Q'—>

Question: *Where are the chimpanzees?*

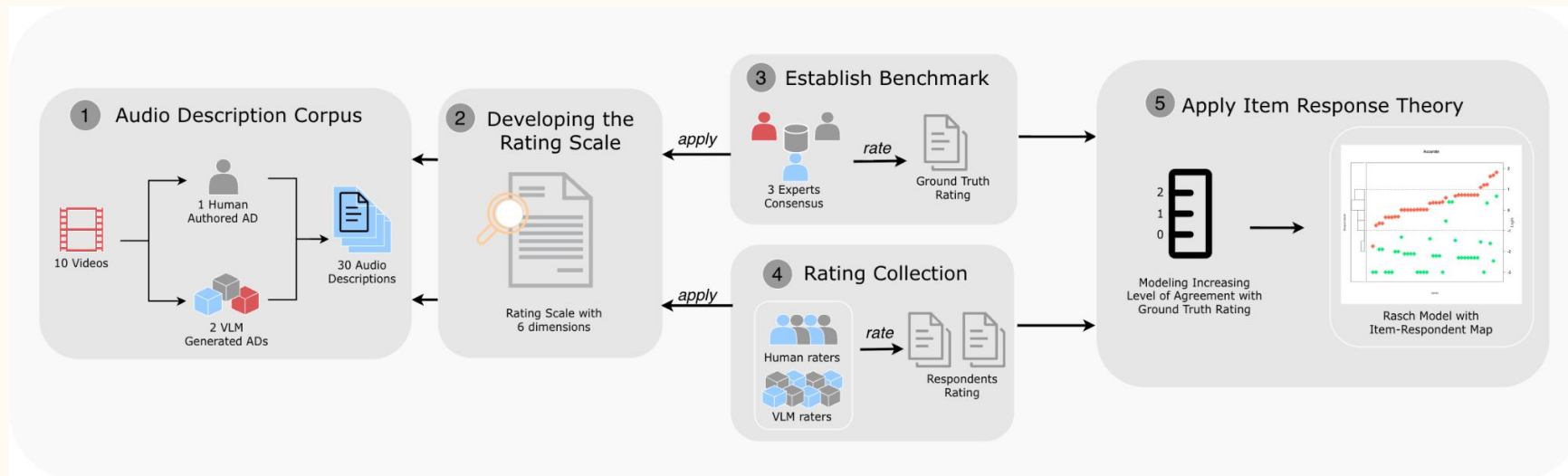
- Answer **Without Strategic Prompting:** They are in a lush forested environment, likely in tropical region such as East Africa
- Answer **With Context Prompting:** They are in a lush forested environment, likely within **Gombe National Park, Tanzania**

Limitations

- GenAD: produces more contextual descriptions, but still struggles with content prioritization and the timing and placement of descriptions
- RefineAD: functional, but we're still refining the interface to give describers a smoother editing experience. RefineAD is designed withscreen reader compatibility, but we need to improve functionality more in order for it to be used by BLV users.
- AdaptAD: builds on prior dialogue-agent work, but we haven't yet tested how all three components integrate together

Evaluate AD quality

- Volunteer and AI-generated AD scale coverage, but quality is not checked in a systematic way
- We formalize a rating scale: six dimensions, four content (Accurate, Prioritized, Consistent, Equal) + two formatting (Method, Timing) and conduct a study to evaluate both human and VLM as raters of AD quality



Evaluate AD quality

Alignment: top VLMs approximate expert ratings at levels comparable to humans, but their justifications are generic and lack the diagnostic value of human feedback

Scale usability: six dimensions proved cognitively demanding to apply consistently, especially for non-expert raters, and we will still need to keep refining the scale more

The Future: Bridging Gaps through Accessibility Education

Colin Stipe, M.A.

95.9% of websites are inaccessible and fewer than 3% of US computer science courses mention disability or accessible design!

- Interested in the impact of exposure to accessible technology on computer science students
- Conducted 60-90 minute, semi-structured interviews with six alumni from SFSU's Computer Science MS program who worked on technical aspects of YouDescribe
- Phenomenological approach to qualitative analysis

The Future: Bridging Gaps through Accessibility Education

5 of the 6 started work on YouDescribe without any particular interest in the project:

- *“We just needed to select a professor to work with. . so I joined”*

None had significant experience with or interest in accessible design before starting work on YouDescribe:

- *“In my four years as an undergrad, I never really heard of accessibility”*
- *“I hadn’t really ever thought about [accessibility]. At my first job [before grad school], I thought of it more like, oh, it’s one of those things I have to check off before I need to share my product.”*

The Future: Bridging Gaps through Accessibility Education

All alumni reported that their experience working on YouDescribe increased their understanding of accessibility and improved their attitude toward creating accessible technologies:

- *“Before working on YouDescribe, I thought providing the facilitations or the help to people with disabilities was something that could only be achieved by a special group. Like hospital rehabilitation. But YouDescribe is a project that let me know that what I’m doing can also help disabled people”*
- *“YouDescribe showed me how important accessibility is. Coming from a completely technical background, we sometimes tend to miss the needs of disabled people, so this project kind of gave me an insight on why we should keep this in mind while developing products.”*

The Future: Bridging Gaps through Accessibility Education

All alumni maintained their original career goals, but 5 of 6 said they now incorporate accessibility into their current work:

- *“At work, whenever I’m approaching building a website, my development phase now has accessibility as one of the tests. So that is something that I, as a software developer, made that change consciously”*

All alumni said that their work on YouDescribe reshaped their view of disability. Without any formal instruction, they expressed key aspects of the social model of disability in everyday terms:

- *“I think it’s just a light cast. There’s no difference as a person. They just need some special help which may not be common, maybe not well provided or widely provided by the society.”*
- *“I don’t see them as disabled. . . because they need more support to succeed, they’ve been ostracized by a society that separates those who are ‘able’ from those who are not. I don’t see that difference anymore.”*

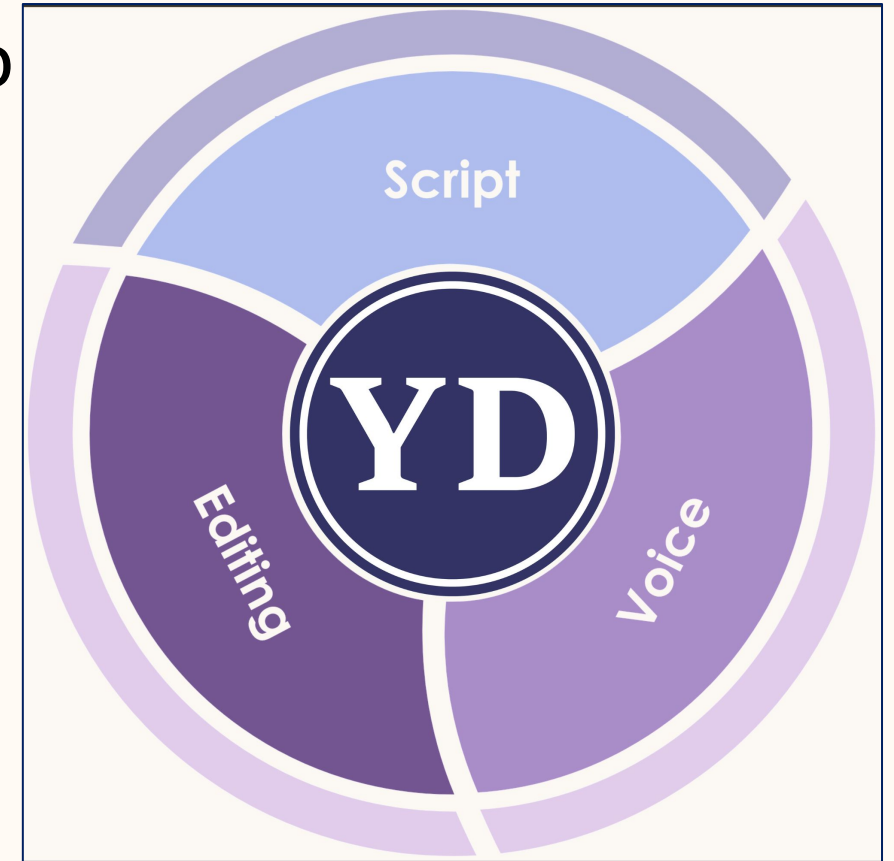
The Future: Advancing ADx3 and AD Quality Control

- Integrate ADx3 end-to-end on YouDescribe, test how GenAD, RefineAD, and AdaptAD work together in practice
- Evaluate with BLV users, validate both the system and the rating scale with the audience they serve
- Refine the quality scale for scalable human use, and explore hybrid human–VLM evaluation workflows
- Close the loop: use quality signals to improve GenAD drafts and guide describers in RefineAD

Evolving Accessibility Since 2013

For 13 years, we've been committed to
breaking barriers.
Now, we're taking the next leap
forward.

New release: May 21, 2026
Celebrating Global Accessibility
Awareness Day



Thank you for your attention!

A special thank you to YouDescribe inventor and mentor Dr Joshua Miele. There is no YouDescribe without you.

This work was supported by grants from:

- NIDLIRR- RERC on Blindness and Low Vision
- Ability Central (formerly The Disabilities Communication Fund)
- NEI
- Department of Education

